Where's Waldo: CS 8803 VLM Final Report

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Abstract

We present WALDO, a lightweight geolocation framework that fine-tunes open-source vision-language models on Google Street View data with LoRA, introduces a locatability-based curriculum, and applies RAG at inference. Our design is motivated by the recent success of retrieval-only pipelines, while targeting applications that require fine-grained localization and may lack the storage or infrastructure for large vector databases. However, our fine-tuned 7–8B models yield only modest gains, if any, over strong baselines. We therefore focus on analyzing the likely sources of this gap, including a loss–task mismatch in which CE-style objectives fail to encode geospatial structure, benchmarking limitations of thresholded accuracy and distribution shift in Flickr-based test sets, tension between open-ended text generation and continuous coordinate prediction, and the role of locatability as a training curriculum versus a data-filtering signal. Code can be found at the repository link below.

https://github.gatech.edu/nvasan7/WALDO

1. Introduction

Geo-localization, the task of inferring the precise coordinate location of an image, is a fundamental open challenge in the field of computer vision . Geo-localization assists with navigation of embodied agents, targeted advertising on social media platforms, open-source intelligence programs in defense, tagging messy or missing data in databases, aiding in search-and-rescue operations and many more application. This problem requires a massive amount of world knowledge and general information, something that vision language models (VLMs) inherently possess through pretraining.

Existing methods for geo-localization using VLMs use a general finetuning approach, a complicated framework and tool set to improve context, or some combination of both. However, existing methods fail to retrieve precise predictions when geo-locating images, potentially due to training on highly-locatable data or over-reliance on niche external tooling to provide context not apparent in the original image [6][10]. Additionally, there are some approaches that utilize a retrieval-augmented-generation (RAG) database with base models, combining the existing strengths of multimodal models with reference images from a vector database to direct the model to a solution [2]. However, performance will always be bottlenecked by the scale of the database. The crux of the problem is the fact that some images simply lack a variety of visual information from which to make an informed guess.

We present WALDO, a lightweight geo-localization framework that combines the improved reasoning capabilities of a fine-tuned VLM with the added precision and context of a RAG database. Instead of disregarding the low-locatable data, we instead reframe the problem as one of difficulty and introduce a locatability-based curriculum to train the model on progressively harder images. In supervised fine-tuning, we fine tune base Qwen models using LoRA with street-view images, utilizing a coordinate-span-weighted cross entropy loss to prioritize higher-digit precision. Finally, at inference time, we integrate RAG to retrieve both the k most and least similar embeddings to ground the predictions.

Our contributions through this work include:

1. We introduce a *lightweight geolocation framework* (WALDO) that combines parameter-efficient fine-tuning (LoRA) with Retrieval-Augmented Generation (RAG), built with a simplified infrastructure to reduce the computational and storage overhead associated with massive retrieval-only paradigms.

2. We propose and implement a *locatability-aware curriculum learning strategy* to stratify training difficulty.

This approach enables the model to effectively learn from a wider distribution of challenging street-view images, unlike prior works that discarded difficult-to-locate images.

3. We develop a novel coordinate-span-weighted Cross-Entropy loss designed to improve the model's sensitivity to geo-spatial proximity during continuous coordinate token generation. We utilize this to analyze and reveal why common *CE objectives are fundamentally insufficient* for guiding fine-tuning processes in continuous coordinate prediction tasks.

2. Related Work

2.1. Existing Geo-localization Frameworks

Recent works have shown great promise in geo-localization tasks, utilizing traditional networks, VLMs, or external tooling to contextualize the street-view images. Geo-CLIP utilizes a CLIP-inspired image-to-GPS retrieval that aligns images with a continuous GPS embedding space via contrastive learning [3]. However, this model relies on mapping the entire planet as a function, encoding any given image and sampling from a distribution to find the best guess. It is unable to leverage any pretrained knowledge or reason through features that a language model backbone could provide. Georeasoner utilizes a Qwen-VL model to finetune using highly locatable street-view images [6]. The model determines a 'locatability score' for each image, employing a Maskformer to generate segmentation masks for different visual classes (e.g. street, sky, building) weighted against their prevalence in an image. GeoReasoner then fine-tunes with only highly locatable data, excluding complex geographies from the tuning dataset. GeoChain is benchmark that leverages CoT reasoning with GeoReasoner's locatability score to evaluate model's abilities to reason their way to a solution [10].

Other frameworks include complex pipelines and tools to support the reasoning capabilities of VLMs. NAVIG added analytical reasoning with external tooling alongside a fine-tuned VLM [12]. Other approaches exclude finetuning completely, arguing in favor of a retrieval only approach to estimate location [2]. Despite these inventive approaches to build on top of a VLM, these tools result in convoluted frameworks that scale poorly and give only slight advantages to their simpler counterparts. For example, NAVIG's training data and external "guidebooks" are heavily based in the GeoGuesser online game, which doesn't translate well to other applications. Ultimately, we seek to define a more effective way to finetune models while limiting tooling. We complement the learning of the model with retrieval to evaluate the importance of both approaches.

2.2. Curriculum Learning

Curriculum learning was introduced by Bengio et. al in 2009 [1] as a method to smooth the learning process and gradually introduce more difficult data. This approach enabled deep, non-convex training objectives to better approximate the global maximum without getting stuck in local minima of the loss surface. Across multiple papers, authors found that on average, Curriculum Learning had a net improvement on the performance of fine-tuning models [8] [11]. This concept has been successfully extended to pretraining and fine-tuning, though some works did mention that these results varied based on the combination of model and dataset, suggesting a need for careful ablation [9]. While many prior works focus on utilizing a vanilla learning curriculum with a linear progression of difficulty, other paradigms like Pacing and Interleaved strategies have also shown to improve performance, demonstrating the importance of matching the proper scheduled learning with the difficulty metric and base models [11].

There were two guiding principles that underline the curriculums of past works: *accuracy* and *scale*. When creating a metric from which to base the curriculum learning, it is crucial to ensure that the difficulty is directly correlated to the training objective. Labels for difficulty can be created by humans or generated with computers, although [9] found favorable results with model-generated labels as these were more consistent than expert human labeling. In terms of scale, the range of difficulty must possess a continuous gradient across the data; if there is a gap or the change in difficulty is too sharp, this could compromise curriculum learning's effectiveness. Therefore, while curriculum learning has shown to work well in fine-tuning language model tasks, it is important to select the proper difficulty metric that correctly scales the learning for a given task, and to trial the approach with different models to find which data/model combinations work best to leverage the employed curriculum.

3. Methods

3.1. Locatability Metric

The difficulty metric we select for our curriculum is GeoReasoner's locatability metric[6]. The core idea behind this metric is that images with a wide variety of highly-locatable features (sky, street, building, etc) should be easier to locate than street-view images with fewer or less locatable class features. The locatability score is calculated using a CLIP-based visual-text pairing network that combines the properties of semantic elements in the image with the importance weights of those weights calculated with the CLIP-embeddings. Waldo utilizes the same weights from GeoReasoner and open-source MaskFormer model to calculate the locatability score for each of it's images for training.

The locatability metric equation is as follows:

$$\text{locatability}(I_{seg}, W_{loc}) = \sum_{k=1}^{n} I_{seg}(k) \cdot W_{loc}(k) \quad (1)$$

In the above equation, I_{seg} is an n-length vector that is the area ratio of each of the n semantic classes identifiable by the segmentation masks of the MaskFormer. W_{loc} is similarly an n-length vector that is the pre-calculated importance weights of each n classes in contributing to the image's overall geo-localability. The weights are calculated over a set of 3K geo-localization textual clues and the semantic segmentation labels using Sentence-BERT. The score itself ranges from 0 to 1, where 0 indicates little to no classes present that make the image locatable, making it difficult to locate. A score near 1 means the image is full of highly locatable semantic features, making it easier to locate.

3.2. Dataset

The finetuning dataset was scraped from the Google Street View API and includes street-view images from across the globe. When scraping, we limit our search to exclude major sections of oceans. We pick a coordinate at random and utilize the Google Maps API to resolve to the nearest available street-view image and associated address. The final data includes an image and a related CSV record with image identifiers such country, region or town names as well as geographic coordinates. To impute missing location identifiers, we apply reverse geocoding to latitude-longitude pairs and map country codes to standardized country and continent names. We fill all null entries using the predicted fields to preserve the original GSV identifiers as much as possible. Finally, the locatability scores are binned into three levels to support curriculum learning according to Table 1. We sample the dataset such that there are a balanced number of images (9816) in each locatability bin, yielding roughly 29,448k images in total.

Score Interval	Locatability Bin
$[0.0, 0.2)$	Hard
$[0.2, 0.4)$	Medium
$[0.4, 1.0]$	Easy

Table 1: Mapping from locatability scores to curriculum bins.

3.3. Supervised Finetuning

Our approach draws on the geolocation framework proposed by [2], as illustrated in Figure 2. However, instead of applying RAG (see Section 3.4) directly to the base model, we fine-tune both Qwen2.5-VL-7B-Instruct and Qwen3-VL-8B-Instruct using locatability-aware supervision on our Google Street View dataset and integrate RAG at inference. Although using RAG alone can be more compute-efficient, performance is often bottlenecked by the scale of the vector database, which can be a limitation in resource-constrained settings. Additionally, retrieval methods alone have not yet been shown to fully resolve fine-grained localization errors at the street or city level.

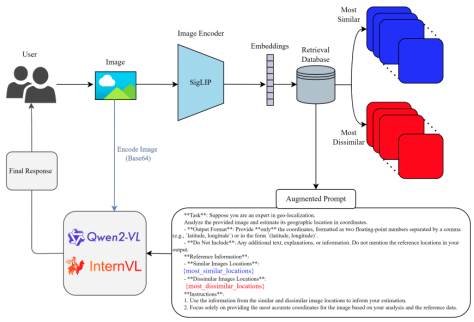


Figure 1: Street Level Localization (Bıçakçı, Y. S. et. al) framework.

Prior works have used the locatability score outlined in Section 3.1 as a filtering strategy, removing images with a score below 0.4 to ensure the model trains on easier-to-locate or higher-quality images. Because this may not reflect real-world image distributions, we instead retain all images and use their associated binned locatability labels. We create a curriculum such that the model trains on progressively more difficult images (e.g. easy $\rightarrow$ medium $\rightarrow$ hard) while conducting cumulative evaluation. Each sample is converted into a multimodal chat-style input (see Appendix 4) and passed into the model along with the image and target coordinates. The model is evaluated on the accuracy of predicted latitude and longitude. The output coordinates are generated and parsed from text rather than using a regression head in order to preserve the model's text-generation capabilities at inference.

We also compare two related loss choices. The baseline loss is standard cross-entropy loss that weights all tokens equally. Equation 2 a coordinate-span-weighted CE loss, where the target is formatted as a tagged coordinate string (e.g., $< coord > \ldots < /coord >$) and the objective simply upweights tokens inside the span while downweighting the rest of the response with a base weight ($w_{\text{base}} = 0.4$). This is meant to encourage instruction-following behavior while penalizing inaccurate coordinate generation more strongly. We treat this lightweight finetuning as a precusor to retrievel-based refinement.

$$\mathcal{L}_{\text{weighted}} = \frac{\sum_{t=1}^{T} w_t \, \ell_t}{\sum_{t=1}^{T} w_t}, \tag{2}$$

$$w_t = \begin{cases} 1.0, & t \in [s, e), \\ w_{\text{base}}, & \text{otherwise}, \end{cases} \tag{3}$$

$$\ell_t = -\log p_\theta(y_t \mid \mathbf{x}, y_{<t}) \, . \tag{4}$$

3.4. Retrieval-Augmented Generation

Similar to prior works, namely G3 [5] and Street-Level Localization [2], we incorporate Retrieval-Augmented Generation (RAG) into our pipeline to enhance the geolocation capabilities of our vision-language models. We construct a large-scale FAISS index database to allow the models to reason over visually similar images with known geographic coordinates. At inference, we retrieve the most similar and dissimilar images for a given query image and feed these into the model to improve location prediction accuracy.

Our RAG database is built in accordance to the Street-Level Localization [2] framework, using a combined set of approximately 10 million globally-distributed natural images. This dataset is formed by merging two sources: the OSV-5M dataset (Open Street View) and the EMP-16 dataset (Earth Media Photography), as visualized in Fig. **??**. Each image in the database is paired with its latitude and longitude coordinates, and indexed using high-dimensional visual embeddings derived from the siglip-so400m-patch14-224 model. This SigLIP encoder was chosen for its strong performance on image retrieval tasks and its compatibility with downstream vision-language prompting pipelines.

At inference time, we embed the test image using the same SigLIP encoder and perform a nearest neighbor search over the FAISS index to retrieve the top-K most similar and bottom-K most dissimilar images where $K = 16$. Their corresponding coordinates are included in a structured multimodal prompt to the model, enabling reasoning over both visual and geographic priors.

3.5. Datasets

To evaluate the many geolocation models explored in this project we chose two datasets, **IM2GPS3k** [4] and **YFCC4k** [7], due to their widespread use in the literature and public availability. Both are globally diverse datasets of natural images retrieved from the popular photography website Flickr, as visualized in Fig. 3. These natural image datasets are primarily used for model evaluation as opposed to street view images because street view datasets are uniform in style and perspective and can be made trivial with RAG. However, they do represent a deparature from the training data distribution.

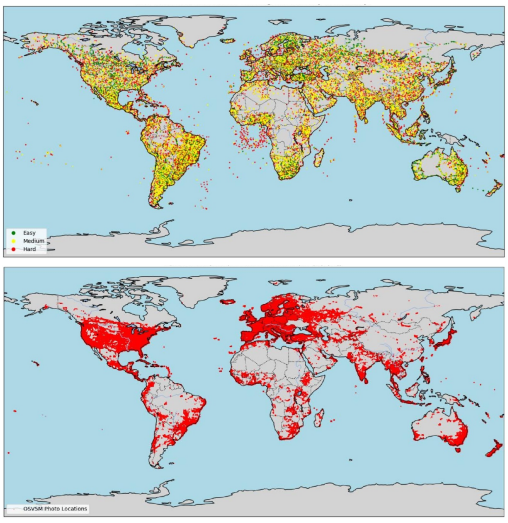


Figure 2: Coverage of the global points for the training dataset (top) vs. coverage of the image locations in the RAG dataset (bottom)



Figure 3: Evaluation Image Samples. From IM2GPS3k (top) and YFCC4k (bottom).

3.6. Evaluation Pipeline

In the evaluation sets of IM2GPS3k and YFCC4k, each sample consists of an image-coordinate pair. The model must determine the latitude and longitude coordinates given only the input image and a small system prompt describing the task [1]. Because the model outputs are the predicted latitude and longitude coordinate, we employ Haversine dis-

[1] See repository for more details.

tance (Eq.5) to obtain the distance in kilometers between the predicted and ground truth coordinates.

$$d = 2r \cdot \arcsin\left(\sqrt{\sin^2\left(\frac{\varphi_2 - \varphi_1}{2}\right) + \cos(\varphi_1)\cos(\varphi_2)\sin^2\left(\frac{\omega_2 - \omega_1}{2}\right)}\right) \tag{5}$$

4. Experiments, Results, Ablations

In the literature, the standard format for evaluating results is done by listing percentages of how many predictions landed within certain distance thresholds. These thresholds are: 1km (Street), 25km (City), 200km (Region), 750km (Country), and 2500km (Continent). We acknowledge that this thresholded approach is an inherently imprecise method for measuring continuous localization accuracy, as it fails to account for variations in real-world geographic boundaries and sizes. However, to maintain alignment with prior work, we report our results in this way.

Our experimentation pipeline involved obtaining these thresholded geolocation accuracy metrics for the chosen baseline vision-language models, our modified versions of these base models, and the various ablation combinations of our augmentations. We compare our method combinations against each other, across different base model variants, and against the reported scores by prior works to paint a complete picture of how the methods perform. The base models we chosen for our variants are Qwen2, Qwen2.5, and Qwen3 at various model sizes. These are reasonable baselines as they represent state-of-the-art generalist VLMs pre-trained on web-scale image-text data without explicit geolocation supervision.

We benchmark both the base models and the finetuned models both with and without RAG to assess the model variants and components of our framework. Tables 2 and 3 contain all key metrics for both the YFCC4k and IM2GPS3k datasets.

4.1. Baselines and Key Results

Overall, the results show modest gains, if any, over the base Qwen model variants. One notable patten is the negative impact of RAG on base models' coarse-grained (e.g. continent, country etc.) localization accuracy when compared to base models without RAG. For example, on IM2GPS3k (Table 3), the Qwen3-VL-8B-Instruct base model's Continent accuracy dropped from 82.48% (No RAG) to 74.61% (RAG). However, we do see small benefits to the base models on the finer-grained localization tasks, such as at the Street level. We hypothesize this could be due to distribution shift between the globally distributed, street-view RAG database and the Flickr-based natural image benchmarks. This may cause the retrieved images' coordinates to provide misleading global context while still offering insight into local visual cues.

Our fine-tuned WALDO models also show little improvement over the base models when RAG is excluded, regardless of the loss type (standard CE versus custom coordinate-span-weighted). Interestingly, the Waldo-Qwen3-VL-8B-Instruct + Coord loss achieved the highest Continent accuracy overall at 62.28% on YFCC4k (Table 2) and 82.68% (Table 3) on IM2GPS3k, outperforming base models and other variants. Furthermore, when RAG was applied the Waldo-Qwen3-VL-8B-Instruct + Coord loss variant achieved the best Street-level accuracy on IM2GPS3k at 8.38%, providing some evidence that a custom loss function may assist with low-level coordinate precision when paired with retrieval. However, the performance is inconsistent across the granularity levels, preventing us from drawing any significant conclusions.

Ultimately, the absolute geolocation accuracy of our variants fall short of most prior works. For instance, on IM2GPS3k (Table 3), the best WALDO Street accuracy of 8.38% is significantly lower than the 24.30% achieved by Street-Level Localization [2]. Similarly, the City accuracy of 28.09% is well below the 38.00% reported by GeoReasoner [6]. This is a substantial performance gap that highlights the fundamental issue with our approach - the cross-entropy-based objective fails to encode the geospatial structure sufficiently for high-precision localization.

4.2. Ablations

Tables 4 and 5 report the various ablations performed across model size and retrieval abilities. The results clearly illustrate the scaling up the base model size is a primary driver of performance, with larger variants generally showing consistent gains across all granularities. However, in IM2GPS3k, the Qwen3-VL-30B-A3B-Instruct model achieved the highest baseline performance without RAG for all granularities except City, outperforming the even larger Qwen2.5-VL-72B-Instruct variant, showing that the enhanced pretraining corpus and training recipes of newer model versions is impactful as well. Additionally, the previously observed negative impact of RAG on coarse-grained localization persists even for larger models, demonstrating that data composition of the vector database cannot be overcome by simply increasing model size alone.

5. Discussion

In this paper we present a lightweight geolocation framework designed to finetune open-source VLMs on street-view imagery using curriculum learning and retrieval methods. We conduct supervised finetuning on smaller, 7-8B Qwen models on Google Street View data and augment

Prior Works	Street	City	Region	Country	Continent
GeoLocSFT	5.21%	18.58%	32.64%	53.10%	72.10%
GeoCLIP	9.59%	19.31%	32.63%	55.0%	74.69%
GeoDecoder	10.3%	24.4%	33.9%	50.0%	68.7%
Street-Level Localization	**24.3%**	**35.3%**	**44.5%**	**59.5%**	**75.2%**
Base model (No RAG)					
Qwen2.5-VL-7B-Instruct	1.54%	10.26%	22.67%	44.04%	61.32%
Qwen3-VL-8B-Instruct	2.56%	10.89%	**22.71%**	42.00%	58.05%
Base model (RAG)					
Qwen2.5-VL-7B-Instruct	2.84%	7.72%	13.65%	30.16%	49.74%
Qwen3-VL-8B-Instruct	3.70%	10.49%	17.55%	35.12%	56.68%
WALDO model (Finetuned)					
Waldo-Qwen2.5-VL-7B-Instruct + CE Loss	1.51%	10.22%	22.59%	**44.06%**	61.35%
Waldo-Qwen3-VL-8B-Instruct + CE Loss	2.49%	**11.24%**	22.68%	42.31%	58.16%
Waldo-Qwen3-VL-8B-Instruct + Coord loss (Ours)	2.34%	10.58%	22.66%	43.63%	**62.28%**
WALDO model (Finetuned + RAG)					
Waldo-Qwen2.5-VL-7B-Instruct + CE Loss	2.84%	7.76%	13.72%	30.18%	49.83%
Waldo-Qwen3-VL-8B-Instruct + CE Loss	**3.75%**	10.54%	18.01%	35.32%	56.26%
Waldo-Qwen3-VL-8B-Instruct + Coord loss (Ours)	3.55%	10.74%	19.80%	37.74%	58.88%

Table 2: **YFCC4k** key results: geolocation accuracy of models at different geographic granularities. Street means the coordinates were within 1 km of the ground truth, City is 25 km, Region is 200 km, Country is 750 km and Continent is 2500 km.

Prior Works	Street	City	Region	Country	Continent
NAVIG-Qwen2-VL-7B	5.5%	28.9%	49.1%	68.3%	84.0%
GEOClip	14.11%	34.47%	50.65%	69.67%	83.82%
GeoReasoner	10.00%	**38.00%**	-	**83.00%**	-
GRE	11.30%	35.30%	**52.70%**	69.30%	**85.70%**
Street-Level Localization	**24.30%**	35.10%	44.50%	59.50%	75.20%
Base model (No RAG)					
Qwen2-VL-7B-Instruct	4.51%	24.75%	44.36%	64.43%	77.92%
Qwen2.5-VL-7B-Instruct	3.96%	23.50%	42.81%	62.32%	76.94%
Qwen3-VL-8B-Instruct	7.04%	**28.67%**	**47.23%**	67.62%	82.48%
Base model (RAG)					
Qwen2-VL-7B-Instruct	5.61%	19.55%	28.53%	44.88%	62.30%
Qwen2.5-VL-7B-Instruct	6.11%	20.25%	29.43%	46.98%	64.83%
Qwen3-VL-8B-Instruct	8.17%	27.36%	38.57%	56.99%	74.61%
WALDO model (Finetuned)					
Waldo-Qwen2.5-VL-7B-Instruct + CE Loss	3.89%	23.47%	42.64%	62.52%	77.04%
Waldo-Qwen3-VL-8B-Instruct + CE Loss	7.21%	28.26%	46.38%	**67.97%**	82.58%
Waldo-Qwen3-VL-8B-Instruct + Coord loss (Ours)	6.41%	27.19%	44.94%	66.60%	**82.68%**
WALDO model (Finetuned + RAG)					
Waldo-Qwen2.5-VL-7B-Instruct + CE Loss	6.14%	20.25%	29.36%	46.85%	64.46%
Waldo-Qwen3-VL-8B-Instruct + CE Loss	8.14%	27.46%	38.71%	57.72%	74.84%
Waldo-Qwen3-VL-8B-Instruct + Coord loss (Ours)	**8.38%**	28.09%	40.17%	59.19%	76.21%

Table 3: **IM2GPS3k** key results: geolocation accuracy of models at different geographic granularities. Street means the coordinates were within 1 km of the ground truth, City is 25 km, Region is 200 km, Country is 750 km and Continent is 2500 km.

Base model (No RAG)	Street	City	Region	Country	Continent
Qwen2-VL-7B-Instruct	1.34%	9.28%	19.75%	38.25%	53.09%
Qwen2-VL-72B-Instruct	3.31%	14.95%	27.82%	44.91%	57.19%
Qwen2.5-VL-7B-Instruct	1.54%	10.26%	22.67%	44.04%	61.32%
Qwen2.5-VL-32B-Instruct	3.20%	13.41%	26.02%	46.70%	66.90%
Qwen2.5-VL-72B-Instruct	**3.84%**	**15.15%**	**27.93%**	46.96%	62.37%
Qwen3-VL-8B-Instruct	2.56%	10.89%	22.71%	42.00%	58.05%
Qwen3-VL-30B-A3B-Instruct	3.42%	13.98%	27.87%	**50.93%**	**67.09%**
Base model (RAG)					
Qwen2-VL-7B-Instruct	3.66%	8.58%	14.64%	30.45%	49.96%
Qwen2-VL-72B-Instruct	5.51%	14.02%	22.71%	41.38%	61.29%
Qwen2.5-VL-7B-Instruct	2.84%	7.72%	13.65%	30.16%	49.74%
Qwen2.5-VL-32B-Instruct	4.76%	13.44%	23.17%	42.67%	62.16%
Qwen2.5-VL-72B-Instruct	**5.89%**	**14.24%**	**24.23%**	**44.25%**	**63.98%**
Qwen3-VL-8B-Instruct	3.70%	10.49%	17.55%	35.12%	56.68%
Qwen3-VL-30B-A3B-Instruct	4.21%	11.77%	19.97%	38.98%	60.76%

Table 4: **YFCC4k** ablations and comparisons: geolocation accuracy of models at different geographic granularities. Street means the coordinates were within 1 km of the ground truth, City is 25 km, Region is 200 km, Country is 750 km and Continent is 2500 km.

Base model (No RAG)	Street	City	Region	Country	Continent
Qwen2-VL-7B-Instruct	4.51%	24.75%	44.36%	64.43%	77.92%
Qwen2-VL-72B-Instruct	7.54%	33.57%	52.05%	68.74%	80.25%
Qwen2.5-VL-7B-Instruct	3.96%	23.50%	42.81%	62.32%	76.94%
Qwen2.5-VL-32B-Instruct	6.91%	30.13%	46.58%	64.93%	79.95%
Qwen2.5-VL-72B-Instruct	7.87%	**34.33%**	51.85%	69.40%	82.25%
Qwen3-VL-8B-Instruct	7.04%	28.67%	47.23%	67.62%	82.48%
Qwen3-VL-30B-A3B-Instruct	**9.28%**	33.80%	**52.75%**	**73.07%**	**86.05%**
Base model (RAG)					
Qwen2-VL-7B-Instruct	5.61%	19.55%	28.53%	44.88%	62.30%
Qwen2-VL-72B-Instruct	8.38%	29.00%	40.47%	57.96%	74.21%
Qwen2.5-VL-7B-Instruct	6.11%	20.25%	29.43%	46.98%	64.83%
Qwen2.5-VL-32B-Instruct	9.31%	30.20%	43.24%	60.36%	77.98%
Qwen2.5-VL-72B-Instruct	**9.54%**	30.16%	41.98%	60.43%	79.31%
Qwen3-VL-8B-Instruct	8.17%	27.36%	38.57%	56.99%	74.61%
Qwen3-VL-30B-A3B-Instruct	9.34%	**30.86%**	**43.91%**	**63.70%**	**79.55%**

Table 5: **IM2GPS3k** ablations and comparisons: geolocation accuracy of models at different geographic granularities. Street means the coordinates were within 1 km of the ground truth, City is 25 km, Region is 200 km, Country is 750 km and Continent is 2500 km.

with a larger RAG database, but still observe subpar performance. In this section, we present our analysis of the likely sources of this gap.

The primary limitation of the WALDO framework is the disparity between the loss optimization used in SFT and and the task objective. Our cross-entropy-based losses are positioned to generate latitude and longitude coordinates as discrete tokens, which fundamentally fails to encode the continuity of geospatial relationships required for image localization. Since the model treats coordinate values like 48.85 and 48.90 as tokens, it has no understanding that a predicted latitude of 48.85 is geographically closer to 48.86 than to 48.90 in coordinate space. We could have used a regression head to predict continuous coordinates but then we override the model's text generation capabilities, preventing integration with retrieval and other downstream multimodal applications and effectively rendering the use of a VLM over a vision-only model obsolete. Ultimately, this structural is-

sue is the reason why our models cannot effectively discern between Paris, Texas and Paris, France for example. By contrast, other successful prior works employ textual Chain-of-Thought reasoning [10] [12] [6] or classification of discretized globe embeddings [5] to force the model to learn these relationships.

There are also several secondary factors that compounded the performance gap. The domain standard for reporting results is thresholded accuracy, which measures whether a prediction falls within predefined distance bounds. This introduces significant imprecision, as mentioned in Section 4. This is further exacerbated by the standard use of Flickr-based natural imagery test sets (YFCC4k and IM2GPS3k) to evaluate street-view-trained models, amplifying distribution shift and making high-precision localization challenging. Another consideration is the role of locatability. Other works use it as a data-filtering signal to improve data quality. The inclusion of hard-to-classify images in our training curriculum could have introduced noise and degraded performance. Lastly, our preliminary results indicate that 7-8B model size is insufficient and was likely a significant bottleneck for model performance.

5.1. Future Work

There are many natural extensions of this work. First, ablating the role of locatability-aware curriculum learning compared to supervised finetuning across a mixture of images should be a priority. Next, compute allowing, different model families and larger models (e.g. 72B+) should be tried since we have shown model size improves performance on average. The SFT dataset can be improved in terms of quality and quantity by increasing geographic diversity as well as adjusting locatability thresholds to allow for less noisy training data.

Perhaps most importantly, however, future work should focus on augmenting this pipeline with a methodology to encode geospatial relationships in some capacity. We believe that chain-of-thought reasoning [6] [12] is a compute-efficient and promising direction that can be done at scale by leveraging a dataset such as GeoChain [10], which includes over a million Mapillary street-view images with 21 QA pairs for each image. A reasoning model placed into our existing framework could enhance the discernment of retrieval context, thereby mitigating the coarse-grained localization failure observed when RAG introduced global noise.

5.2. Ethical Implications

Although this work aims to step towards solving an open problem in computer vision, there are some major ethical concerns for a large scale implementation of this work. First, there is a privacy concern as any exterior image could potentially be used for geo-localization, leading to doxxing or unauthorized surveillance of these individuals. Additionally, the availability of data is biased towards more developed nations, providing poor performance in areas with lower amounts of publicly available imagery. Further, since many of the large-scale datasets that would be useful in training are owned and operated by big tech companies, this leads to an imbalance of utilization. It should be noted that visages and personally identifiable information was pre-scraped out of the available data utilized in the training of our model.

6. Work Division

See Table 6 [2].

7. Appendix

7.1. Prompts

System Prompt

You are an expert in image geolocation. Your task is to analyze images and predict the GPS coordinates (latitude, longitude) of where the image was taken. Base your geolocation on visual clues such as architecture style and building materials, vegetation and climate indicators, road signs, license plates, and language on signage. However, you must ONLY output the structured format. Do NOT provide explanations, reasoning, or descriptions.

User Prompt

Provide ONLY the GPS coordinates in decimal degrees using this EXACT format:
`<coord>(+12.34567, -123.45678)</coord>`
No extra text.

Figure 4: Prompting protocol used across training and inference.

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Student Name	Contributed Aspects	Details
Nikhita Vasan	Model Training and Infrastructure	Primarily responsible for SFT using LoRA. Designed and built the end-to-end training infrastructure, including data collation, training pipelines, code modularization, custom loss functions, and curriculum implementation. Led root-cause analysis to diagnose unsuccessful training runs.
Ethan Haarer	SFT Dataset Curation + CL Research	Developed data generation pipeline - scraped 40k images from Google Street View API, adapted locatability scoring code to dataset and conducted research on curriculum learning for this use case.
William Stevens	Model Training, RAG and Experimentation	Primarily responsible for RAG development and experimentation. Assisted in developing model training code, loss functions and parallelization. Replicated RAG framework from [2], ran all model variants and displayed results in report / presentation.

Table 6: Contributions of team members.

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